

Cross-Document Dependencies for Persistent Native Memory

Separating Interaction Selection from Evidence Consumption

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Abstract

Persistent native memory keeps retrieved records independently addressable and reusable, whereas ordinary retrieval-augmented generation (RAG) serializes them into one causal prefix. A matched 150-question audit shows why this distinction is budget dependent. Near 512 selected source tokens, packed RAG and independent Progressive Retrieval Attention (PRA) are tied within uncertainty (.427 versus .433 official score). Near 1,024 tokens, packed RAG rises to .527 while independent PRA falls to .333. This paper asks which inter-record dependencies explain that transition and whether they can be selected and executed sparsely.

We separate two problems: choosing useful cross-document dependencies and presenting their consequences in a form the frozen decoder can consume. For selection, we instrument the unchanged host attention path, extract a graph over physical layer/head/token edges, and replay exact sparse masks under frozen retrieval receipts. On a ten-question mechanism cohort, 0.1% of physical edges reaches .1488 F1 versus .1490 for packed RAG, although the absolute gap is small. On 150 frozen test questions, task-aware pair-NLL ranking at 1% reaches .1243 F1 versus .1148 for raw attention and .1119 for packed RAG. Its paired interval versus packed includes zero, and every frontier is non-monotonic: useful sparse headroom exists, but attention magnitude is not a dependable causal ranking.

For evidence consumption, we test 221 bounded cross-document retrieval policies. The selected dense policy gains 2.94 support-span points on validation but only 0.89 on frozen test, where 95.3% of admitted tokens are distractors. Expansion-only native generation lowers F1 from .0896 to .0761. More decisively, a gold-support expansion also lowers F1 when its spans remain independently contextualized. A matched 2×2 control shows that pair attention alone is neutral near 512 tokens, while applying it after expansion recovers 84.1% of the expansion-induced F1 loss; the factorial interaction estimate is positive but uncertain. Near 1,024 tokens, full-pair and boundary-only attention recover 37.1% and 41.8% of the packed F1 deficit. Both paired F1 intervals exclude zero, but neither method reaches packed quality. Thus finding dependencies and making their effects usable are distinct causal requirements. The current mechanisms remain research controls rather than deployment defaults.

1 Introduction

RAG systems solve a retrieval problem and a context-construction problem. Retrieval selects useful evidence. Context construction then decides how that evidence is presented to the language model. The common implementation packs retrieved text into a causal sequence,

$$D_1 D_2 \cdots D_m Q,$$

and freshly prefills the whole sequence. This creates a dense and order-dependent computation graph: D_2 can attend to D_1 , D_3 can attend to both D_1 and D_2 , and the final query can attend to all preceding states.

PRA exposes a different design point. Each D_i can remain a persistent addressable record with reusable model-native K/V. The query can later select and materialize those records without rebuilding all source states. The advantage is straightforward when a selection repeats or when records recur across changing queries. The harder question is composition: independent records do not contain the cross-document causal history introduced by ordinary packed RAG.

Paper 3.2 localized this difference and then reconciled its earlier endpoint against Paper 3.3 under one selector, cohort, prompt, evaluator, and source-token budget. Near 512 tokens, packed RAG and independent PRA are statistically tied. Near 1,024, packed causal composition gains while independent realization degrades. Contiguous transport remains exact, pre-RoPE rebinding preserves packed coordinates, and independent records track a packed control with document-to-document attention disabled. The unresolved question is therefore not whether packing is universally better, but when cross-document dependencies become useful as selected context grows.

These results motivate the paper’s central question:

Which cross-document dependencies are useful, and how must their effects be executed so that a frozen decoder can consume them?

We therefore separate three quantities:

1. **persistent source state**, stored once as reusable PRA records;
2. **request-specific interaction selection**, which determines which records/tokens/layers need contextualization;
3. **request-specific interaction execution**, which applies a small amount of the frozen base model’s own computation.

The intended architecture is

persistent PRA K/V $\rightarrow$ sparse interaction selector
 $\rightarrow$ small request-local cross-record computation $\rightarrow Q$.

Contributions. This study makes six contributions:

1. **Physical causal-interaction instrumentation.** Exact replay over layer, head, record-pair, and token edges without changing model weights.
2. **Sparse causal headroom.** Very small edge subsets can approach packed behavior on mechanism cohorts, although the frontier is unstable.
3. **Attention magnitude is not causal utility.** Pair-level leave-one-group-out targets rank useful interactions better directionally than raw attention mass on the powered test.
4. **Bounded cross-document retrieval expansion.** Lexical, dense, hybrid, and oracle policies measure how much omitted support can be recovered under explicit native-K/V budgets.
5. **Evidence exposure is insufficient.** Even gold support fails to improve answers when added records remain independently contextualized.
6. **Linked neural interaction repairs consumption failure.** Pair and boundary self-attention recover expansion-induced degradation near 512 tokens and part of the packed deficit near 1,024, while leaving the deployment gate closed because gains are incomplete and metric dependent.

Table 1: Matched Paper 3.2/3.3 audit. Δ is independent PRA minus packed RAG; intervals are paired 10,000-sample bootstrap intervals. Budgets count source whitespace tokens.

Cohort / budget	n	Packed Off.	PRA Off.	Δ Off. [95% CI]	Packed F1	PRA F1
Paper 3.3 test, 512	150	0.427	0.433	+0.007 [-0.073,+0.087]	0.086	0.090
Paper 3.3 test, 1,024	150	0.527	0.333	-0.193 [-0.280,-0.107]	0.087	0.057
Paper 3.2 eval, 512	30	0.433	0.333	-0.100 [-0.267,+0.067]	0.085	0.081
Paper 3.2 eval, 1,024	30	0.533	0.333	-0.200 [-0.333,-0.067]	0.101	0.080
Paper 3.2 legacy-v2	30	0.567	0.500	-0.067 [-0.300,+0.167]	0.102	0.103

2 Related Work

RAG and retrieval-conditioned language models place external evidence in the model context or retrieve it during generation [5, 4, 3]. Sparse-attention systems reduce long-sequence cost through fixed, random, global, or learned access patterns [1, 10, 7]. Other work retrieves external states at attention time or restricts K/V traffic [2, 8, 6]. These approaches motivate sparse access, but the present question is different: the retrieved records are already selected and persist as independently encoded native memory. We ask which request-specific cross-record interactions must be restored to recover the behavior of a conventional packed prefix. CacheBlend also studies reuse when separately cached chunks need request-time fusion [9]; our unit of intervention is an auditable physical attention edge under a frozen RAG selection.

3 Relationship to Prior PRA Work

3.1 Paper 1.5: request-time RoPE binding

Paper 1.5 separated coordinate correctness from causal contextualization and showed that, for RoPE models,

$$K_{\text{post}}(p) = R_p K_{\text{pre}}.$$

Pre-RoPE storage therefore allows a persistent key to be rebound to a request-specific effective position without re-encoding the source.

Paper 3.3 assumes that positional contract and does not reopen the positional transport question except as a runtime qualification requirement.

3.2 Paper 3.0: materialization is a computational choice

Paper 3.0 showed that more materialized K/V is not automatically better and that useful detail depends on the query, evidence distribution, and consumer layer. This motivates a query-conditioned interaction policy rather than a fixed global cross-document mask.

3.3 Paper 3.2: causal localization of the composition gap

Paper 3.2 supplies the direct starting point, but its original 30-question endpoint and Paper 3.3’s later 150-question endpoint used different chunking, retrieval revisions, and budgets. A protocol-locked audit reruns both cohorts with Qwen3-1.7B-4bit, deterministic BM25-v2, 128-token chunks, the same strong reranker, and one frozen receipt per packed/PRA pair. Table 1 reports the resulting budget transition.

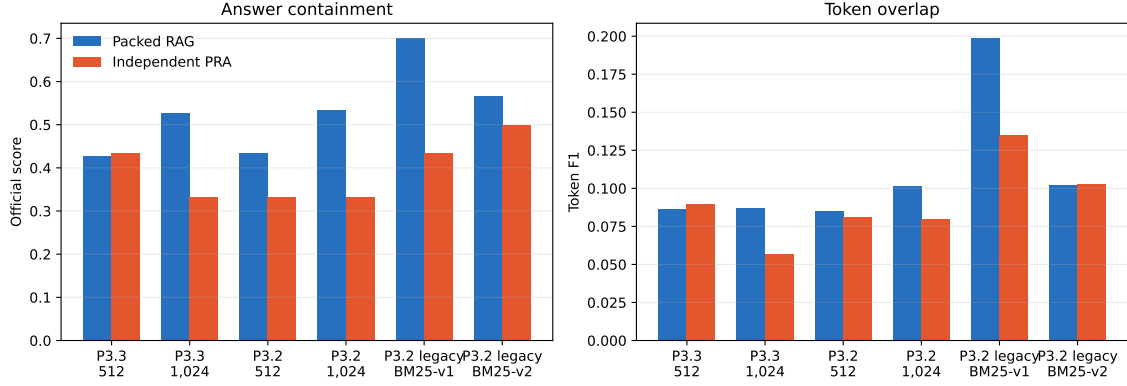


Figure 1: Protocol reconciliation from Paper 3.2. Independent realization and packing are tied near 512 selected source tokens, whereas packed causal composition becomes advantageous near 1,024 under the same selection protocol.

On the 150 Paper 3.3 identities, the 512-token endpoint is a parity result: official score differs by $+.007$ with interval $[-.073, .087]$, and F1 by $+.003$ with interval $[-.010, .017]$. At 1,024 tokens, packed official score rises from $.427$ to $.527$ while independent PRA falls from $.433$ to $.333$. The independent-minus-packed official gap is then $-.193$ with interval $[-.280, -.107]$, and the F1 gap is $-.030$ with interval $[-.046, -.015]$. The earlier $.700$ versus $.433$ comparison is therefore not a general endpoint; its legacy geometry and BM25-v1 selection contributed materially to the gap.

More importantly, Paper 3.2 compared

A = full causal packed RAG,

B = packed RAG with document-to-document attention disabled,

and

C = independent pre-RoPE PRA records rebound to the same positions.

B and C were behaviorally close, while a shape-matched storage/rebinding control was exact across the tested precision modes. The principal semantic difference is therefore cross-document causal interaction.

Paper 3.3 does not attempt to make independent records reproduce every dense packed state. It asks which subset of those interactions is needed for task quality, and whether retrieving omitted evidence can substitute for that neural interaction. The matched audit makes this a scaling question: the target is the dependency structure that emerges between 512 and 1,024 selected source tokens.

4 Problem Formulation

Let the selected evidence consist of m persistent records

$$\mathcal{D} = \{D_1, \dots, D_m\},$$

with total token count T . Let $\mathcal{E}_{\text{full}}$ denote all document-to-document attention edges allowed by ordinary packed causal RAG.

We seek a request-specific sparse edge set

$$\mathcal{E}_q \subset \mathcal{E}_{\text{full}}$$

such that

$$|\mathcal{E}_q| \ll |\mathcal{E}_{\text{full}}|$$

while preserving task utility:

$$U(\mathcal{D}, Q, \mathcal{E}_q) \approx U(\mathcal{D}, Q, \mathcal{E}_{\text{full}}).$$

The system must additionally preserve most persistent native state:

$$C_{\text{new}}(\mathcal{E}_q) \ll C_{\text{prefill}}(\mathcal{D}).$$

Our primary quality target is to recover at least 80% of the independent-to-packed RAG gap under a request-specific interaction budget of at most 10% of dense cross-document interaction.

5 Part I: Selecting Useful Dependencies

5.1 Teacher graph extraction

We instrument packed RAG with frozen retrieval, reranking, selected records, order, token budget, model, and generation settings. For every causal document-to-document attention interaction we record sufficient metadata to construct a sparse teacher graph:

- layer and head;
- source and target record/chunk;
- source and target token positions;
- attention mass;
- distance to document boundaries;
- retrieval rank and score.

Let E be the set of causal token pairs whose source and target belong to different records. The stored score array has shape $[L, H, |E|]$, not $[L, H, T, T]$. Source and target token indices and record identities are stored once, so within-record and causally impossible entries never enter the artifact. For each layer, the observer computes probabilities on a diagnostic side path; the hidden-state update still uses MLX’s host scaled-dot-product-attention kernel. A parity receipt checks the observer path against ordinary host execution before the graph is interpreted.

5.2 Oracle budgets

We first retain the highest-attention edges at budgets

$$0, 0.01, 0.05, 0.1, 0.5, 1, 2, 5, 10, 100\%.$$

A complementary cumulative-mass condition retains the smallest edge set covering

$$50, 75, 90, 95, 99\%$$

of cross-document attention mass.

For replay, a plan contains a Boolean array of shape $[L, H, |E|]$. At layer ℓ , selected physical edges are added to a packed mask that otherwise blocks cross-record attention. The frozen host model then executes the modified mask. The 100% plan must reproduce the packed teacher exactly; the 0% plan must reproduce the blocked packed control. These endpoint checks distinguish a sparsification result from an instrumentation or transport error.

5.3 Interventional ranking targets

Attention magnitude is observational: a large coefficient shows that an interaction occurred, not that it improved the answer. We therefore add a leave-one-group-out utility for a group g ,

$$u_{\text{NLL}}(g) = \mathcal{L}_{\text{answer}}(M_{\setminus g}) - \mathcal{L}_{\text{answer}}(M),$$

where M is the packed teacher and $M_{\setminus g}$ removes only the group’s cross-document edges. A positive value means that removing the group raises gold-answer NLL. We also measure first-step Jensen–Shannon divergence, $u_{\text{JS}}(g)$, between $M_{\setminus g}$ and M .

The hierarchy is document pair $\rightarrow$ layer $\rightarrow$ layer/head $\rightarrow$ token edge. At each tested level, groups are ordered by measured utility and physical edges within a group are refined by teacher attention. A second pair-level proxy ranks by

$$\max(0, u_{\text{NLL}}(g)) a_e,$$

where a_e is the teacher attention of edge e in group g . This is a finite-difference-times-attention diagnostic, not a gradient attribution. These rankings are compared with raw attention at matched physical-edge budgets 0, .01, .05, .1, .5, and 1%; 100% remains an endpoint parity check.

Every natural question is sampled only after restricting the dataset to a named frozen Paper 3.3 split. Atomic per-question checkpoints bind the dataset, model, reranker, split digest, ranking targets, and budgets, so a resumed run cannot mix configurations. A ten-question validation diagnostic localizes the .1% plans by layer, head, and document pair and selects ranking targets. The powered confirmation then requires at least 100 previously untouched test questions. Learned selection remains locked unless an interventional test ranking meets the absolute quality gate and its F1 frontier is non-decreasing through 1%.

5.4 Oracle gate

The learned-selector program proceeds only if a sparse oracle can approach packed-RAG quality. The initial gate is

$$\text{F1} \geq .19, \quad \text{official} \geq .67,$$

or statistical indistinguishability from packed RAG while using no more than 5% of dense cross-document interactions.

6 Where Useful Interactions Live

We analyze oracle-selected interactions by:

- transformer layer;
- attention head;
- source/target retrieval rank;
- document pair;
- chunk pair;
- token boundary distance;

- query overlap;
- entity/token overlap;
- gist similarity.

This analysis determines the appropriate routing granularity. If useful interactions are largely pair-level, we avoid token-level prediction. If they concentrate in a few layers or heads, request-local compute can be reduced further.

7 Learned Interaction Selection

This section specifies the next-stage design; it is not evaluated in the present paper. Training is deliberately locked by the failed mechanism gate.

7.1 Pair selector

The first learned policy predicts interaction importance for selected pair (D_i, D_j) :

$$s_{ij} = f_{\theta}(g_q, g_i, g_j, r_i, r_j, \phi_{ij}),$$

where g_q is a query gist, g_i, g_j are persistent record/chunk gists, r_i, r_j are retrieval/reranker features, and ϕ_{ij} contains pairwise semantic features.

The pair selector is trained against oracle pair importance while controlling a request-specific interaction budget.

7.2 Layer/head selector

If the oracle is concentrated, we extend the policy to

$$s_{ij\ell h}$$

or a factored approximation.

The paper prefers the simplest selector that reaches the quality/economics gate.

8 Sparse Request-Local Contextualization

Synthetic parameter-free K/V append failed in Paper 3.2. The present paper therefore prioritizes on-manifold computation using the frozen base transformer’s own projections and attention operations.

Candidate mechanisms are:

Sparse packed-mask composition. Use selected interactions in a request-local composition pass as the first oracle/runtime control.

Boundary interaction. Permit selected real boundary tokens to interact across selected record pairs.

Selected-token interaction. Use a learned token gate within selected record pairs.

Selected-layer propagation. Apply interaction only at selected layers, then propagate corrected request states through the remaining decoder.

Persistent record K/V remains immutable. All corrections are request-local.

9 Part II: Evidence Exposure and Consumption

Physical-edge selection asks which neural interactions to replay. A cheaper alternative asks an earlier question: whether one selected record identifies a small, previously unselected interval in another selected record. Let $S_i \subset D_i$ be the frozen first-stage span for record D_i . For every allowed ordered pair $i \neq j$, we issue either

$$S_i \longrightarrow \text{search}(D_j) \quad \text{or} \quad (Q, S_i) \longrightarrow \text{search}(D_j).$$

The first form measures selected-span evidence alone; the second uses the user query to suppress associations unrelated to the current reasoning path. Search never escapes the records authorized by the frozen first-stage receipt.

We implement exact chunk BM25, frozen dense similarity, reciprocal-rank fusion, an entity/keyterm channel, and a support-span oracle. Powered dense runs use BAAI BGE-base-en-v1.5 with its query and document encodings kept distinct; the dependency-free signed-hash encoder is an SDK fallback and is not labeled as a semantic result. Dense vectors for authorized peer chunks, the request, and selected source spans are prepared once per request. We report that preparation separately from scorer latency so dense and fusion rows cannot inherit order-dependent warm-cache advantages. For lexical rank $r_{\text{lex}}(c)$ and dense rank $r_{\text{dense}}(c)$, the primary hybrid score is

$$s_{\text{RRF}}(c) = \frac{1}{60 + r_{\text{lex}}(c)} + \frac{1}{60 + r_{\text{dense}}(c)}.$$

RRF avoids calibrating unlike raw scores. A normalized weighted fusion is kept as a secondary research arm. We also compare symmetric all-pairs expansion, higher-to-lower and lower-to-higher retrieval rank, a top-record hub, and pairs that pass a query-relevance threshold.

The runtime, rather than the search policy, owns safety and accounting. It rejects unauthorized targets, removes overlap with the initial selection and other expansions in logical source coordinates, clips the final interval to a global token ceiling, and distinguishes resident from newly encoded native K/V. For candidate set C and admitted source intervals A , it records

$$B_{\text{cross}} \in \{32, 64, 128, 256, 512\}, \quad k_{\text{pair}} \in \{1, 2, 4, 8\},$$

together with requested, deduplicated, reused, and newly materialized tokens. The hash-addressed **CrossDocumentExpansionReceipt** contains source and target identities, offsets, channel scores, ranks, keyterms, policy revision, and failure labels, but not raw source text.

Expansion and neural interaction remain separate interventions. *Expansion only* appends selected native source intervals without cross-record SA. *Pair SA* admits base-model attention only between retrieval-linked record pairs. *Boundary SA* further limits each linked pair to a source suffix and target prefix. Finally, a diagnostic physical-edge arm ranks teacher edges only inside retrieval-linked pairs. These masks execute through the unchanged host attention implementation, so their edge fractions describe logical sparsity, not kernel acceleration.

To identify which intervention matters, we also project the same discovered source–target record links onto the original selected spans and replay pair SA without adding evidence. This gives the

Table 2: Best bounded cross-document expansion point per channel on the 150-question validation split. Gains are absolute percentage points over the same frozen first-stage selection. “Q” indicates query conditioning.

Channel	Q	Span gain	Chunk gain	Extra K/V	Distractor	Search ms
Lexical	yes	2.67	3.71	12.66%	90.00%	337.8
Dense	no	2.94	4.27	12.66%	86.00%	258.7
RRF hybrid	yes	2.44	3.79	12.66%	88.00%	344.7
Weighted hybrid	yes	1.83	4.35	12.66%	87.33%	340.5
Entity/keyterm	yes	1.72	2.71	12.66%	92.67%	337.0
Oracle	yes	23.17	30.55	19.65%	0.00%	338.2

factorial design

	no cross-record SA	pair SA
original evidence	independent PRA	pair-SA only
expanded evidence	expansion only	expansion + pair SA.

A matched boundary-only arm restricts interaction to eight tokens at each record boundary. The same frozen test identities are replayed at 512 and 1,024 source tokens. This separates evidence acquisition from neural consumption and ties the interaction study directly to the budget transition in Figure 1.

The public abstraction is fail-closed. Parameter-free modes have stable configuration and plugin contracts, while oracle and weighted modes require an explicit experiment flag. Runtime authorization, deduplication, budget enforcement, and receipt generation also wrap custom plugins. No mode becomes an SDK default from a smoke result: qualification requires a powered frozen cohort, five-seed uncertainty, at least two model sizes, bounded quality and latency, and preferably cross-family replication.

The first-stage selection cache stores tokenizer-independent source intervals and checks the candidate receipt, selector identity, token ceiling, and record limit before replay. Model tokenization occurs only during native encoding. An interprocess receipt check during this work exposed two defects in the historical BM25-v1 implementation: corpus term frequency had been accumulated where document occurrence was required, and floating sums traversed an unordered query-term set. BM25-v2 counts each term once per document and sorts query terms before accumulation. We invalidated and regenerated all preliminary expansion results after this correction; every new receipt carries the v2 revision. Earlier physical-edge results remain tied to their pinned historical commit and are not silently relabeled as v2 comparisons.

9.1 Frozen validation frontier

We evaluate 221 policy configurations on 150 frozen validation questions. The grid crosses five parameter-free channels and an oracle with selected-only or query-conditioned probes, $k \in \{1, 2, 4, 8\}$, and extra-native-token ceilings from 32 to 512. Table 2 reports the best point per channel subject to an extra-native fraction of at most 20%. These are validation choices, not test results. All non-oracle optima use a 64-token ceiling, which is 12.66% of the initial 506.2-token native selection on average.

Dense selected-only is the predeclared test policy: symmetric chunk search, $k = 1$, and a 64-token global ceiling. It raises supporting-span coverage from 41.33% to 44.28% and gold-chunk recall from 38.57% to 42.85%. Five repeated paired bootstrap calculations place the span gain in a conservative

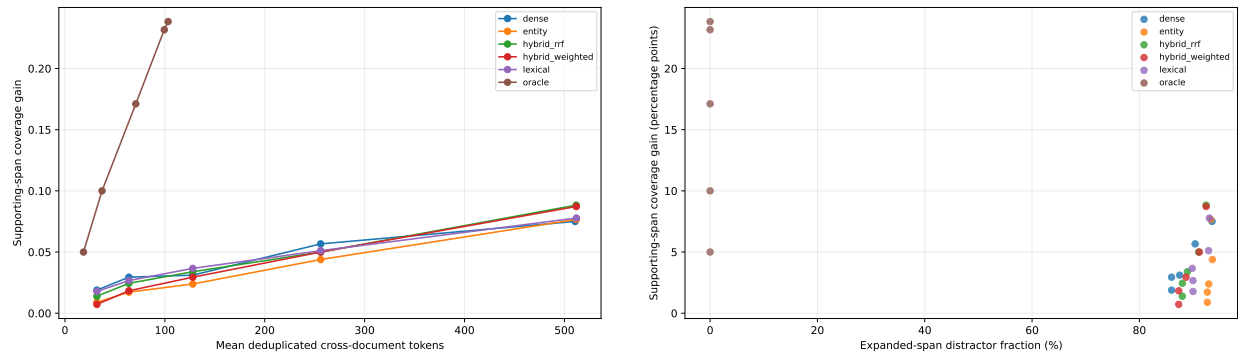


Figure 2: Validation expansion frontier. Left: support-span gain against additional native K/V. Right: gain against the fraction of admitted tokens outside annotated support. Oracle points expose substantial unresolved selection headroom.

95% interval of [1.39, 4.72] points and the chunk-recall gain in [2.51, 6.22] points. Answer-string availability rises by 2.67 points, with interval [0.67, 5.33].

The positive retrieval effect is not yet an efficiency or answer-quality win. The selected policy requests 1,521.1 cross tokens per question before overlap removal and global clipping, but admits only 64.0: a 23.77-fold request-to-materialization amplification. Moreover, 86% of admitted tokens do not overlap annotated support. The compact oracle gains 23.17 span points with 99.4 extra tokens and no distractors; dense expansion therefore recovers only 12.71% of oracle span headroom. Figure 2 makes both the bounded gain and the unresolved selectivity gap visible. Semantic vector preparation adds 358.7 ms per request in this exhaustive research implementation and is reported separately from the 258.7 ms dense search.

Direction and interval controls sharpen the diagnosis. Symmetric lexical expansion gains 2.67 span points, whereas the query-selected-pair-only rule gains 1.33 points while reducing the distractor fraction from 90.0% to 38.7% and the extra-native fraction from 12.66% to 5.37%. Restricting deduplication to logical source intervals raises lexical chunk-recall gain to 6.96 points but does not close the oracle gap; the corresponding oracle gains 28.36 points with only 5.58% extra native tokens. These controls show that pair restriction and interval accounting can make expansion cleaner, but the current parameter-free score still does not identify most missing support.

9.2 Frozen test retrieval

We replay the validation-selected dense policy unchanged on the disjoint 150-question test split. Table 3 includes two oracle controls but does not use them to retune the policy. The locked policy raises supporting-span coverage by 0.89 points, gold-chunk recall by 1.63 points, and answer-string availability by 1.33 points. Conservative paired bootstrap intervals are [0.22, 1.78], [0.52, 3.08], and [0.00, 3.33] points, respectively. Thus the evidence-recovery effect transfers, but the answer-availability interval touches zero.

The matched 64-token oracle gains 9.83 span points and 25.93 chunk-recall points while using fewer physical tokens than the dense policy. Dense selected-only therefore recovers 9.04% of matched-budget oracle span headroom. Its 95.33% distractor fraction also exceeds validation’s 86.0%. This generalization gap keeps the parameter-free expansion policy **RESEARCH_ONLY**; the following generation experiment asks whether the small retrieval gain nevertheless changes answer behavior.

Table 3: Frozen test retrieval. Oracle rows diagnose available headroom and do not participate in policy selection.

Condition	Span cov.	Chunk recall	Answer avail.	Extra K/V	Distractor	Tokens
No expansion	40.83	38.54	60.00	0.00%	—	0.0
Dense, 64 (locked)	41.72	40.18	61.33	12.65%	95.33%	64.0
Oracle, 64	50.67	64.48	60.00	9.00%	0.00%	45.6
Oracle, 256	63.72	70.07	60.00	21.37%	0.00%	108.1

Table 4: Frozen 150-question Qwen3-1.7B native generation replay. Lower NLL is better; times are per question in milliseconds. Linked masks execute a dense host kernel and are mechanism controls, not sparse-speed results.

Condition	F1	Official	Gold NLL	Encode	TTFT	Total
Packed RAG	.0863	.4267	5.3106	305.1	68.0	207.9
Independent PRA	.0896	.4333	3.0310	362.2	75.0	214.1
Expansion only	.0761	.3733	2.6310	423.0	76.3	219.0
Linked pair SA	.0874	.4400	3.0117	808.2	71.4	214.8
Linked boundary SA	.0879	.4467	3.0152	776.2	68.2	212.1
Linked top-attention	.0783	.3800	2.6583	775.5	68.2	211.6

9.3 Native answer replay

Table 4 replays the frozen test selection through Qwen3-1.7B-4bit. The model sees 84.3 additional model-native tokens on average after the 64 source-token intervals are tokenized. Expansion-only lowers F1 from .0896 for independent PRA to .0761 and official score from .4333 to .3733. Its paired F1 effect is $-.0135$ with conservative 95% interval $[-.0272, .00005]$; the official-score effect is $-.0600$ with interval $[-.1400, .0200]$. The interval endpoints prevent a directional significance claim, but the means do not support promotion.

Retrieval-linked pair SA reduces the expansion-only F1 deficit relative to independent PRA by 84.1%; the boundary variant reduces it by 87.6%. Their remaining paired F1 effects, $-.0021$ and $-.0017$, have intervals $[-.0099, .0056]$ and $[-.0111, .0081]$. Both linked modes are therefore statistically compatible with independent PRA on this cohort, not superior to it. The apparently favorable expansion-only gold NLL is especially instructive: it improves by .4000 while decoded F1 and official score decline. Answer-likelihood distillation alone would select the wrong deployment policy.

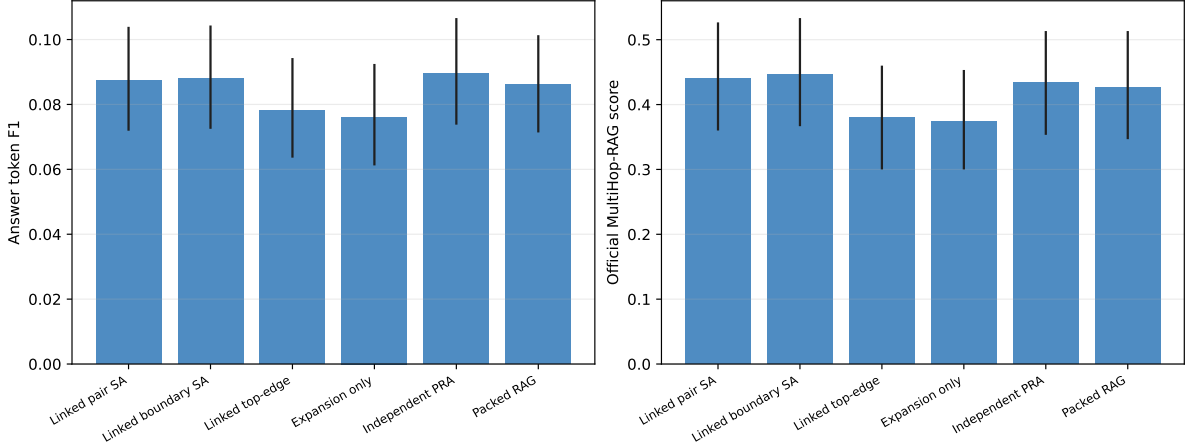


Figure 3: Absolute answer F1 and official MultiHop-RAG score for the frozen Qwen3-1.7B replay. Error bars are conservative envelopes across five paired bootstrap calculations; they do not represent independently trained models.

Gold-support expansion oracle. The matched 64-source-token oracle separates routing error from consumption. It adds 60.1 model-native tokens on average and raises retrieval support-span coverage by 9.83 points, yet expansion-only F1 falls from .0896 to .0829 ($-.0066$, 95% interval $[-.0170, .0037]$) and official score falls from .4333 to .4000 ($-.0333$, interval $[-.0933, .0267]$). Linked pair SA restores F1 to .0888 and boundary SA to .0895; both overlap independent PRA. The packed endpoint is already slightly below independent PRA on this cohort, so a packed-gap-recovery ratio is not meaningful. More importantly, even correctly identified missing support does not improve absolute answer quality when it is merely appended as independently encoded native memory. The mechanism gate is therefore closed for both parameter-free and oracle expansion-only execution.

9.4 Factorial control and budget transition

The preceding expansion result still combines two interventions: admitting additional evidence and allowing retrieval-linked records to interact. We therefore replay the same 150 frozen identities in a 2×2 design at the 512-source-token budget. The interaction-only cell projects the fixed dense router’s source–target links onto the original selected records; it introduces no new span. Table 5 shows the four cells.

Pair SA alone differs from independent PRA by only $+.0021$ F1 ($[-.0081, .0126]$) and $+.0267$ official score ($[-.0333, .0867]$). Expansion alone instead changes F1 by $-.0135$ and official score by $-.0600$. Adding pair SA after expansion restores F1 from .0761 to .0874, or 84.1% of that loss. The difference-in-differences is $+.0092$ F1 with interval $[-.0090, .0280]$ and $+.0400$ official score with interval $[-.0600, .1333]$. The direction supports an interaction role, but the intervals do not establish a factorial effect.

We next change only the frozen source-token budget and replay packed, independent, full-pair, and boundary-pair conditions. Both runs use the same model revision, 150 identities, operating-system image, and package versions; the selection receipt necessarily changes because the 1,024-token condition contains more source evidence. Table 6 and Figures 4–5 report the result.

At 1,024 tokens, packed RAG exceeds independent PRA by $+.0301$ F1 ($[.0145, .0462]$) and $+.1933$ official score ($[.1000, .2800]$). Full-pair SA recovers $+.0112$ F1 over independent PRA ($[.0016, .0220]$), or 37.1% of the packed deficit, while selecting 3.69% of physical edges. Boundary-only SA recovers

Table 5: Matched expansion-by-interaction factorial at 512 selected source tokens. Pair SA uses the same discovered record links with or without admitted evidence.

Condition	Added evidence	Linked SA	F1	Official
Independent PRA	no	no	0.0896	0.4333
Pair SA only	no	yes	0.0917	0.4600
Expansion only	yes	no	0.0761	0.3733
Expansion + pair SA	yes	yes	0.0874	0.4400

Table 6: Cross-document interaction at matched 512- and 1,024-source-token budgets. “Physical edges” is the selected layer/head/token edge fraction; the implementation still executes dense kernels.

Budget	Condition	n	Physical edges	F1	Official
512	Packed RAG	150	0.00%	0.0863	0.4267
512	Independent PRA	150	0.00%	0.0896	0.4333
512	Pair SA only	150	16.86%	0.0917	0.4600
512	Boundary SA only	150	0.04%	0.0816	0.3867
1,024	Packed RAG	150	0.00%	0.0868	0.5267
1,024	Independent PRA	150	0.00%	0.0567	0.3333
1,024	Pair SA only	150	3.69%	0.0679	0.3867
1,024	Boundary SA only	150	0.01%	0.0693	0.3400

+0.0126 F1 ([.0023, .0240]), or 41.8%, with only .0083% of edges. Its official-score change is only +.0067 ([−.0400, .0533]); full-pair SA changes official score by +.0533 with an interval touching zero. Sparse interaction therefore recovers a measurable part of the newly visible F1 gap, not packed equivalence. It also remains a logical sparsity result because the prototype has no sparse attention kernel.

A secondary 512-token replay on a different Apple host produces the same positive factorial direction but a modest endpoint shift. We retain that aggregate and provenance summary as a robustness artifact rather than pooling it with the same-host budget comparison. This sensitivity reinforces the decision to keep linked interaction outside the deployment default.

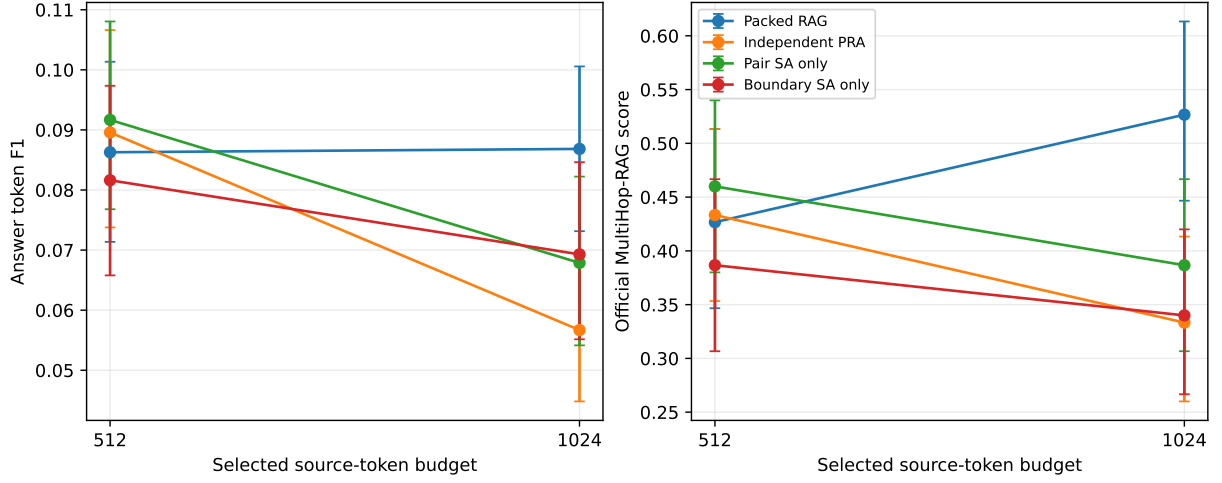


Figure 4: Absolute answer quality across the context-budget transition. Error bars are conservative envelopes over five paired bootstrap calculations, not five independently trained models.

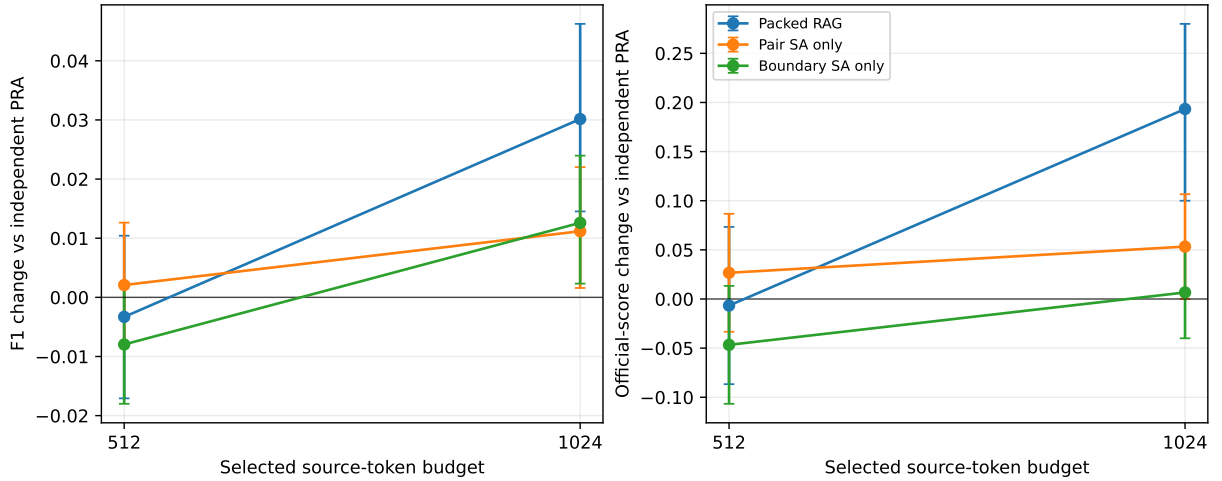


Figure 5: Paired effects relative to independent PRA. Interaction is neutral or negative near 512 tokens; near 1,024, both interaction modes improve F1, but neither closes the packed gap and the official-score evidence is weaker.

9.5 Task-aware region-layer validation

The 1,024-token factorial motivates a finer signal than one scalar per document pair. On ten validation questions, we therefore cross four contiguous decoder- layer bands with equal-width prefix, middle, and suffix windows on both sides of each retrieval-linked record pair. The resulting 36 cells have matched geometry. Per-example gold-answer NLL ranks singleton cells against a packed no-cross-document control. We decode the best singleton and, separately, the union of the four best positive singletons. This is a gold-answer oracle headroom audit, not selector training.

Table 7: Task-aware validation consumption at 1,024 source tokens. NLL gain, F1 change, and official-score change are paired against no-cross packed; higher NLL gain is better. Intervals are conservative bootstrap envelopes over ten questions.

Condition	NLL gain	F1 change	Official change
Best singleton	.1548 [.0748, .2416]	−.0010 [−.0029, 0]	.0000 [0, 0]
Ranked four-cell union	.0941 [.0058, .1918]	−.0023 [−.0048, 0]	.0000 [0, 0]

The singleton’s measured NLL gain exactly matches its ranking signal, but does not transfer to answer F1. The union makes the separation sharper: its summed singleton prediction is +.4382 NLL, while only +.0941 is realized. Useful local effects are therefore strongly non-additive. The singleton and union use .0082% and .0328% of physical edges on average. No suffix-to-prefix boundary cell is selected as the best singleton on any question. In layers 0–6, the boundary cell itself has mean NLL gain −.4308 with interval [−.9197, −.0891], and its contrast against the eight equal-cost geometries is −.3344 with interval [−.6836, −.0731]. The other boundary-band contrasts are negative or near zero.

This development cohort does not reproduce the powered test’s packed advantage, so it cannot estimate deficit recovery. It does answer the validation gate: token-region-by-layer NLL localization exposes local causal sensitivity, but neither a boundary-specific rule nor improved answer consumption. A powered test and learned-selector training remain locked. The next mechanism study must target composition or calibration on validation identities. All masks in this audit still execute through the dense host kernel; timings support no sparse-runtime claim.

9.6 Reduced scale and family replication

We replay the identical selection-cache digest on 30 matched test questions with Qwen3-4B, Qwen3-8B, and Llama-3.1-8B-Instruct. These reduced cohorts test directional transfer; they do not have the power of the 150-question 1.7B run. Table 8 reports paired percentage-point effects against each model’s independent-PRA condition.

Table 8: Model ladder effects versus independent PRA. Columns report F1 changes for the three interventions and the expansion-only official-score change.

Model	<i>n</i>	Expansion F1	Pair-SA F1	Boundary F1	Expansion official
Qwen3-1.7B	150	−1.35	−0.21	−0.17	−6.00
Qwen3-4B	30	+0.64	−1.31	−0.65	−3.33
Qwen3-8B	30	−0.03	+0.92	+0.81	−3.33
Llama-3.1-8B	30	+2.78	−3.39	−4.85	+3.33

No intervention has a consistent sign across scale and family. The only F1 interval excluding

zero is the negative Qwen3-4B pair-SA effect, $[-2.83, -0.09]$ points. Llama expansion-only has the largest positive mean, but its interval is $[-8.42, 13.73]$ points; the corresponding official-score interval is $[0, 10]$ points. Qwen3-8B pair SA is directionally positive but also uncertain. These results reject a universal larger-model consumption claim and keep both expansion and linked attention outside the SDK default surface.

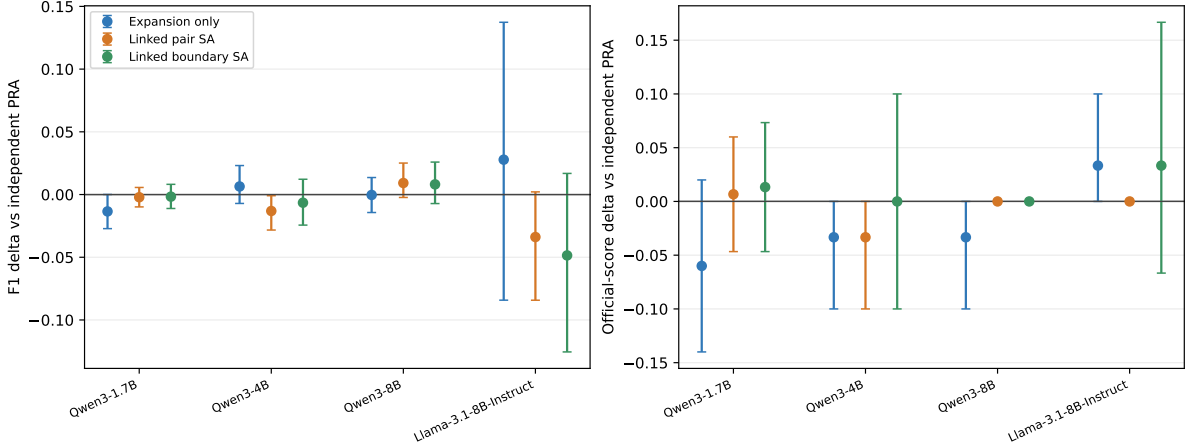


Figure 6: Paired generation effects across model scale and family. Error bars are conservative bootstrap envelopes. The 1.7B point uses 150 questions; the three qualification points use the same 30-question subset.

Qualification decision. The implementation is reproducible and auditable, but neither dense expansion nor linked SA has a stable answer-quality advantage. The stable public behavior therefore remains `mode=none`; expansion, oracle, learned, and recursive policies remain research-only. Appendix A gives the complete predeclared gate and runtime contract.

10 Training Objective

The base model remains frozen initially.

We optimize

$$\mathcal{L} = \lambda_{\text{task}} \mathcal{L}_{\text{answer}} + \lambda_{\text{seq}} KL(P_{\text{sparse}} \| P_{\text{packed}}) + \lambda_{\text{sel}} \mathcal{L}_{\text{selector}} + \lambda_{\text{cost}} C_{\text{interaction}}.$$

Task quality receives the dominant weight. Global K/V reconstruction is not a primary objective because Paper 3.2 showed that state proximity, answer NLL, and generated-answer quality need not agree.

11 Order Robustness

Packed RAG changes its causal graph when selected documents are permuted. PRA records offer the possibility of more stable set-like evidence composition.

We evaluate canonical, reverse, and random permutations under:

- packed RAG;
- independent PRA;

- oracle sparse composition;
- learned sparse composition.

We report task quality together with mean pairwise JS, unique outputs per question, and F1/NLL variance. Lower order sensitivity is considered a benefit only at matched or improved task quality.

12 Experimental Protocol

12.1 Dataset

MultiHop-RAG is the first mandatory dataset because Paper 3.2 already provides frozen retrieval, canonical records, support labels, and powered natural baselines.

Additional datasets are introduced only after the mechanism passes the primary gate.

12.2 Selection

Use the strongest qualified matched selector from Paper 3.2:

$$\text{BM25} \rightarrow \text{BGE-v2-M3}.$$

Candidate and selection receipts are frozen before composition conditions run.

12.3 Training split

The frozen split uses:

- 2,000 training questions;
- 150 validation questions;
- 150 fixed held-out test questions; paired uncertainty uses bootstrap seeds 11, 23, 37, 71, and 101, which are repeated resampling calculations rather than independent model runs.

The split seed is 3301. All 30 identities declared by the final Paper 3.2 rank-8 residual manifest are excluded from every Paper 3.3 split, rather than only from training. The dataset revision, corpus revision, exact identities, legacy manifest, and split digest are persisted. Oracle smoke questions do not train or select parameters.

12.4 Model ladder

M0 Qwen3-1.7B: full discovery/training matrix.

M1 Qwen3-4B: reduced qualification.

M2 Qwen3-8B: reduced qualification.

M3 Llama-3.1-8B: cross-family qualification.

13 Baselines

Mandatory baselines:

1. packed RAG;
2. independent pre-RoPE PRA;
3. Paper 3.2 rank-8 residual;
4. fixed sparse boundary mask;
5. oracle sparse interaction;
6. learned sparse selector/composer.

14 Metrics

Task quality. Official score, token F1, exact match, gold-answer NLL, and answer log-probability.

Distributional fidelity. First-step JS/KL, logit differences, and generated-output agreement.

Sparsity. Selected document pairs, layers, heads, tokens, and total cross-document interaction edges.

Systems. Persistent K/V reuse, newly computed states, request-local FLOPs, composition latency, TTFT, total latency, and memory.

Robustness. Five-seed intervals, order sensitivity, model-scale transfer, family transfer, and precision sensitivity.

15 Causal Dependency Results

15.1 Mechanism scope and endpoint validity

We ran the complete oracle grid on ten deterministically selected MultiHop-RAG questions with Qwen3-1.7B-4bit on an Apple M4 Pro with 48 GiB unified memory. BM25 produced 20 candidates, BGE-v2-M3 reranked them, and a 512-token budget retained three or four records. The model, reranker, corpus, and dataset revisions are pinned in the manifest. This is a one-seed mechanism cohort, not a population estimate. A post-freeze provenance check places all ten identities in the Paper 3.3 training partition, with no overlap with the Paper 3.2 final evaluation identities. The validation diagnostic and powered test confirmation therefore remain disjoint from this mechanism result.

Observer instrumentation matched ordinary host execution on all 10 questions. The 100% physical-edge plan also matched the instrumented packed teacher on all 10, with zero K/V delta and zero first-step JS. The 0% plan matched the packed no-cross-document mask. The average teacher graph contained 38.04 million physical cross-document edges over 28 layers and 16 heads. These checks close the implementation gate: differences inside the sweep are caused by the edge mask, not by an alternate attention kernel.

15.2 Quality–sparsity frontier

Table 9: Paper 3.3 oracle smoke. “Edges” is the fraction of physical layer/head/token-pair cross-document edges. Attention mass is measured under the packed teacher.

Condition	F1	Official	Gold NLL	Edges	Mass
Packed host / oracle 100%	.1490	.600	4.780	100%	100%
Independent PRA	.1431	.600	2.550	0%	0%
Oracle top 0.01%	.1449	.600	2.579	.01%	4.0%
Oracle top 0.05%	.1416	.600	2.571	.05%	19.2%
Oracle top 0.1%	.1488	.600	2.610	.1%	36.2%
Oracle top 0.5%	.0968	.400	4.794	.5%	80.3%
Oracle top 5%	.0990	.400	4.851	5%	91.9%
Oracle top 10%	.1212	.500	4.852	10%	95.1%
Mass oracle 50%	.1403	.500	2.753	.149%	50%
Mass oracle 75%	.1145	.500	3.757	.324%	75%
Mass oracle 90%	.0768	.400	4.758	3.466%	90%
Mass oracle 99%	.1308	.500	4.855	30.902%	99%

At 0.1% physical edges, the oracle recovers 97.3% of this ten-question cohort’s small 0.0059 F1 packed–independent gap: F1 is .1488 versus .1490 for packed RAG and .1431 for independent PRA. Official score remained .600 in all three conditions, so no official-score gap existed to recover. The 0.1% plan did not reproduce the same strings as packed RAG: output agreement was 0/10 and first-step JS remained .110. The result is therefore task-level headroom, not behavioral equivalence, and the relative percentage should not be read as a large absolute quality gain.

The frontier is strongly non-monotonic. Increasing the top-edge budget from 0.1% to 0.5% reduced F1 from .1488 to .0968 even though retained teacher mass rose from 36.2% to 80.3%. Likewise, the smallest set carrying 50% of teacher mass used only .149% of physical edges but did not outperform independent PRA. Raw attention mass is thus an inadequate causal-utility target by itself.

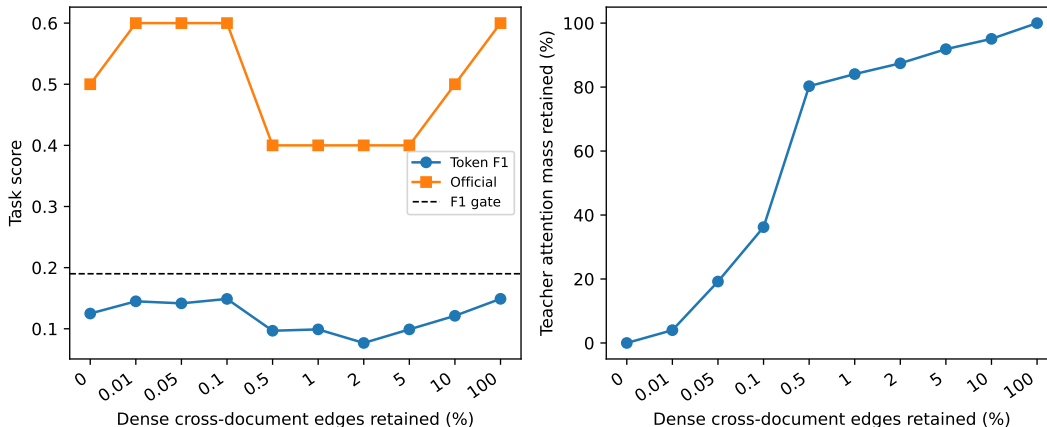


Figure 7: Task quality and retained teacher mass across physical top-edge budgets. The curve is a mechanism smoke over ten questions; connecting lines do not imply monotonicity or statistical precision.

15.3 Interaction localization

Cross-document attention mass was distributed across the network: layers 0–8 contained 27.4%, layers 9–18 contained 31.7%, and layers 19–27 contained 40.8%. No single layer exceeded 4.85% on average, and the largest layer/head cell averaged only .357%. The graph is therefore late-leaning but not localized to one layer or head. Pair mass was more concentrated: the first record supplied 49.1% of mass into the second and 38.8% into the third. These values partly reflect causal geometry and token counts; they are routing features to test, not evidence that early-ranked records are causally sufficient.

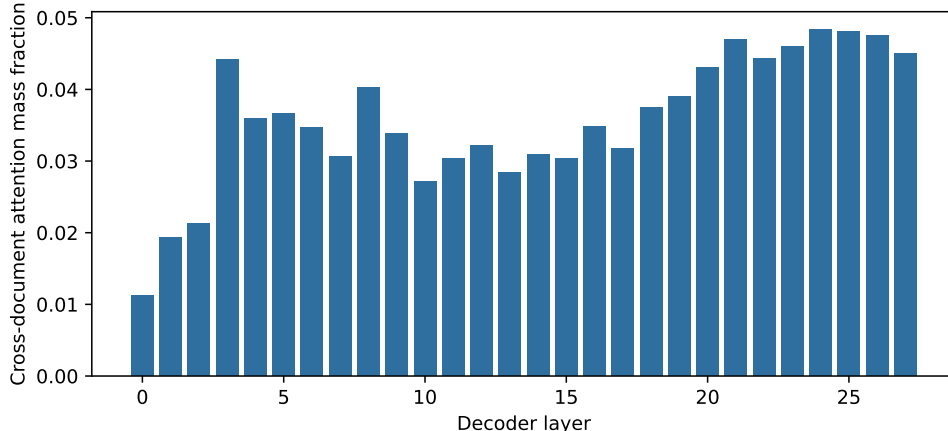


Figure 8: Mean share of packed-teacher cross-document attention mass by decoder layer. Later layers carry more aggregate mass, but no narrow layer band dominates.

15.4 Interventional validation

We next ran the six ranking targets on ten frozen validation questions, using the same model, selector, 512-token budget, and physical-edge replay. The BGE reranker ran on MPS; this execution device is part of the run configuration and receipt. All ten ordinary/instrumented host comparisons and all ten 100% replay checks passed exactly.

Table 10: Frozen validation target selection. Each oracle row reports its highest-F1 budget at or below 1%; it is not a test-set estimate.

Condition	Edges	F1	Official	Gold NLL
Packed teacher	100%	.1094	.600	5.911
Independent PRA	0%	.1673	.500	3.211
Raw attention	.1%	.1706	.500	3.697
Pair NLL	1%	.1659	.600	3.929
Pair JS	.5%	.1727	.600	3.666
Pair NLL $\times$ attention	.1%	.1379	.400	3.861
Layer NLL	.01%	.1540	.400	3.204
Layer JS	.5%	.1525	.500	3.739

This cohort reverses the small-cohort endpoint: independent PRA exceeds packed F1 by .0579, although packed retains .100 more official score. It therefore does not define a packed advantage that

sparse edges can meaningfully recover. The apparent improvements are also uncertain. Relative to the packed teacher, pair-JS at .5% changes F1 by +.0633 (paired bootstrap 95% interval $[-.0067, .1618]$) with zero mean official-score change; raw attention at .1% changes F1 by +.0613 ($[-.0275, .1604]$) while changing official score by $-.100$. Both intervals include zero. Gold NLL ranks conditions differently from generated-answer metrics, so lower teacher-forced loss is not treated as proof of better task behavior.

No ranking has a non-decreasing F1 frontier through 1%. The result therefore strengthens, rather than resolves, the earlier warning: raw magnitude is not a causal utility signal, and coarse leave-one-group-out NLL or first-step JS is not sufficient by itself. The powered test carries raw attention, pair NLL, pair JS, and pair-NLL-times-attention forward. Layer-only policies are dropped after validation because they do not improve the quality frontier.

15.5 Powered frozen-test confirmation

We then evaluate all 150 previously untouched test identities. The model is Qwen3-1.7B-4bit on an Apple M5 with 16 GiB unified memory; the BGE reranker runs on MPS. All 150 ordinary/instrumented comparisons and all 150 full-replay comparisons pass exactly. The run takes 18,486 seconds and records 4,800 condition rows. Table 11 reports the highest-F1 point at or below 1% for each ranking. These are test-set summaries selected over a small prespecified budget grid, not independently tuned deployment settings.

Table 11: Powered 150-question frozen test. Oracle rows show the best F1 budget at or below 1% physical cross-document edges.

Condition	Edges	F1	Official	Gold NLL
Packed teacher	100%	.1119	.3333	5.526
No-cross-document packed	0%	.1105	.3600	3.097
Independent PRA	0%	.1129	.3667	3.114
Raw attention	.05%	.1148	.3867	3.100
Pair JS	1%	.1205	.3800	3.817
Pair NLL	1%	.1243	.3933	3.675
Pair NLL $\times$ attention	.1%	.1215	.4000	3.319

Pair NLL at 1% changes F1 by +.0124 relative to packed RAG (paired bootstrap 95% interval $[-.0055, .0315]$) and official score by +.0600 ($[-.0133, .1333]$). The combined proxy at .1% changes F1 by +.0096 ($[-.0078, .0289]$) and official score by +.0667 ($[-.0067, .1400]$). Pair JS at 1% and raw attention at .05% also have intervals spanning zero. The powered result therefore distinguishes mean frontier shape from established superiority: causal grouping produces the better sparse means, but no task-quality effect is conclusive.

The endpoint ordering is itself informative. Packed F1 differs from independent PRA by only $-.0010$, and its paired interval includes zero. Gold NLL strongly prefers the blocked and independent conditions, so NLL divergence from packed does not translate monotonically into generated-answer quality. This cohort cannot support a claim that dense packed contextualization is a better task endpoint than a sparse oracle merely needs to recover.

The powered .1% plans also reveal different geometries. Raw attention assigns 32.6% of edges to layers 23–25 and 27 and 94.3% to pairs (0, 1) and (0, 2). Pair NLL spreads 84.8% across (0, 2), (1, 2), and (0, 1) and emphasizes early layers 1–2 alongside distributed later layers. Pair JS assigns 86.7% to (0, 1), while the combined proxy assigns 91.7% across the first three pairs and leans later. No leading layer–head cell receives more than .86% of a policy’s selected edges. These are policy-induced

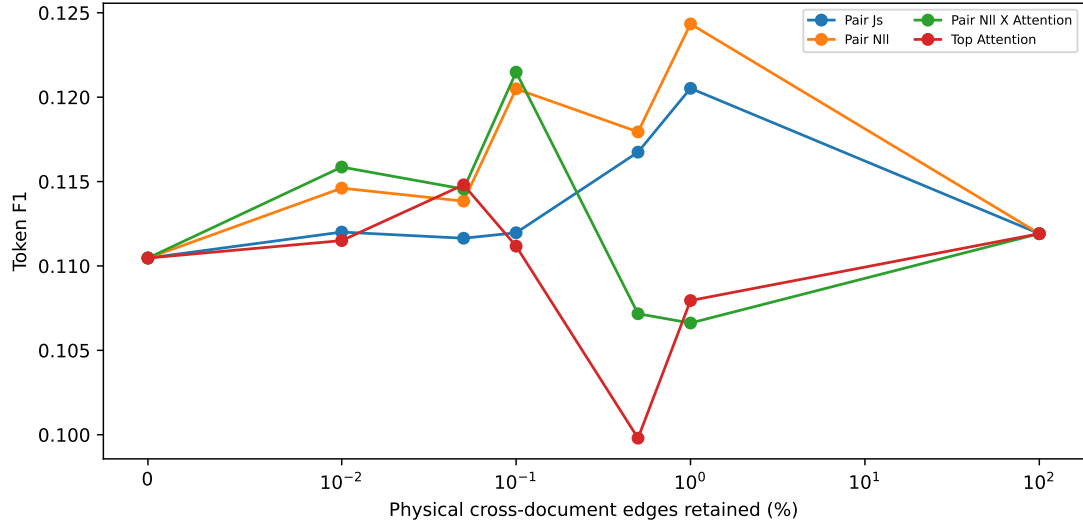


Figure 9: Powered-test F1 frontiers for observational and interventional rankings. Causal pair rankings improve the mean sparse frontier, but all curves remain non-monotonic; lines aid reading and do not imply interpolation.

concentrations, not universal heads, but they show that the causal target materially changes the proposed sparse computation.

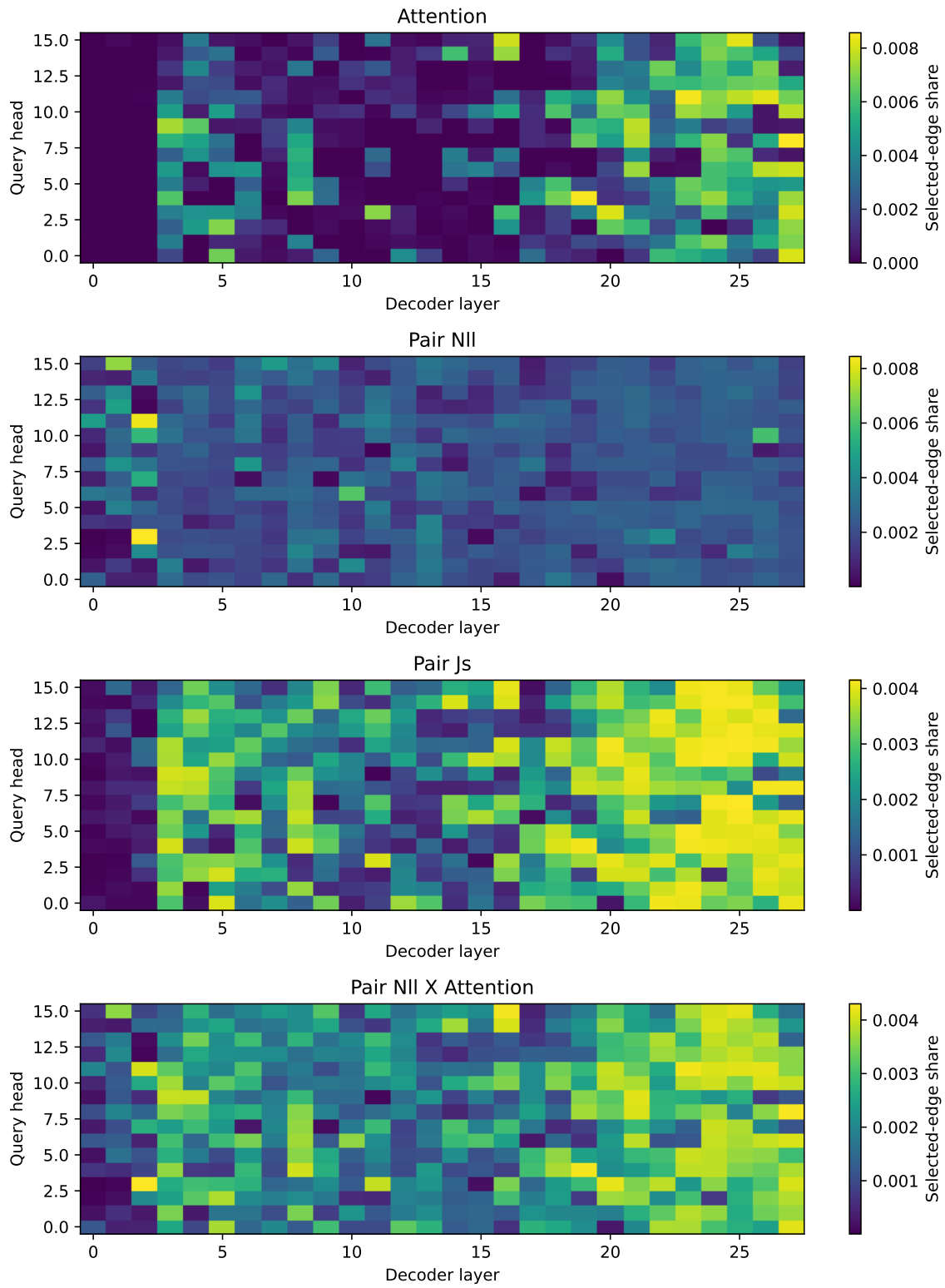


Figure 10: Layer-head allocation of each .1% plan over 150 frozen test questions. Color is selected-edge share within a policy.

15.6 Go/no-go decision

The powered gate is **locked**. The cohort-size requirement is met, but no condition reaches .19 F1 and .67 official score, no frontier is non-decreasing through 1%, and the paired task-quality intervals include zero. Learned selector training therefore remains disabled. The next useful selector target must improve causal localization beyond one scalar utility per document pair; training the current target would turn an unresolved oracle into a learned approximation of an unstable policy.

The reported encoding times, roughly 0.5–0.9 seconds per condition, execute a dense host attention kernel under a sparse mask. Edge fraction is therefore a logical interaction budget, not a measured wall-time or FLOP reduction. A production sparse kernel is downstream work and should be evaluated only after the quality gate passes.

16 Success Criteria

The predeclared practical gate is

$$\text{F1} \geq .19, \quad \text{official} \geq .67,$$

or statistical indistinguishability from packed RAG, while using no more than 5% of dense request-specific cross-document interaction. Approximate 80% gap recovery remains a useful descriptive quantity, but does not replace these absolute thresholds.

Neither the ten-question smoke nor the 150-question frozen test meets these absolute criteria. The smoke’s 0.1% point meets the relative F1-recovery target only because its packed and independent endpoints are close. On the powered test, causal rankings improve sparse mean quality but do not establish a monotonic frontier or a nonzero paired task-quality effect. We do not substitute either post-hoc observation for the prespecified gate.

Retrieval-guided expansion also misses its separate promotion gate. The locked policy recovers evidence at a bounded token cost, but expansion-only generation does not recover 50% of a packed-RAG F1 gap or avoid a negative 1.7B mean. The gold-support oracle shows that improving the router alone is insufficient.

A stronger result matches packed RAG within uncertainty while preserving:

- persistent native K/V reuse;
- lower order sensitivity;
- materially lower request-specific compute.

17 Falsification

Scientifically useful negative outcomes include:

- an oracle requires dense interaction;
- learned selection fails to approach the oracle;
- sparse state propagation destroys quality;
- selector overhead erases saved prefill;
- order robustness disappears at matched quality;
- the mechanism does not transfer across models/families.

18 Discussion

The 0.1% result suggests that conventional RAG may perform substantially redundant document-to-document computation. It does not yet establish that claim: the sample is small, the absolute gate failed, and the present sparse mask still executes a dense kernel. PRA could eventually retain persistent source state and pay only for a small request-specific neural “join” among selected resources, but that economic claim requires both powered quality and a real sparse execution path.

Retrieval-guided expansion gives a more concrete decomposition. Its parameter-free router finds a statistically positive but small amount of missing support, with poor precision. Replacing it with a support oracle substantially improves evidence recovery but still does not improve expansion-only answers. Retrieval-linked pair or boundary attention then restores much of the lost quality. The direct factorial sharpens the claim: pair SA alone is neutral near 512 tokens, and its positive interaction with expansion remains uncertain. At 1,024 tokens, however, both pair and boundary SA produce positive paired F1 effects. Boundary SA obtains the larger mean recovery with far fewer logical edges, but barely changes official score. The sequence locates two distinct research problems: selecting useful peer intervals and contextualizing those intervals without rebuilding a dense packed prefix. Neither can be inferred from gold NLL alone, and the model ladder shows that neither has a model-independent sign.

Taken together, the two experiment families support a narrower claim than “more retrieval” or “more attention”:

useful composition = dependency selection + decoder-consumable execution.

The physical-edge study varies the first term while holding evidence fixed. The expansion study varies evidence exposure, then restores a bounded form of the second term. Raw attention is a weak proxy for the first; independently appended gold support is insufficient for the second.

This would change the role of PRA from a cache/retrieval mechanism into a composable native-memory substrate:

persistent records + sparse request-local contextualization.

If it fails, the result would instead bound how separable transformer source context really is: packed causal contextualization may be too distributed to recover cheaply after independent encoding.

19 Conclusion

The matched audit establishes a context-scale transition: independent PRA and packed RAG are tied near 512 selected source tokens, while packed causal composition is significantly stronger near 1,024. Paper 3.3 targets the inter-record dependencies behind that transition. Exact endpoint checks show that the observer and physical-edge masks isolate this variable. The ten-question smoke finds a promising 0.1% point but also a sharply non-monotonic frontier, so attention mass alone is not a reliable selector target. The 150-question frozen test confirms all endpoints and finds that document-pair causal rankings improve mean sparse quality over raw magnitude: pair NLL reaches .1243 F1 at 1% edges versus .1119 packed. The paired F1 interval nevertheless includes zero, the packed endpoint is not stronger than independent PRA at this 512-token budget, every frontier remains non-monotonic, and the absolute gate is missed. Sparse causal headroom is credible; a qualified selector is not. The finer validation-only token-region-by-layer oracle reduces answer NLL but does not improve F1, its singleton utilities compose poorly, and the suffix-to-prefix boundary

is not preferred. Learned-selector training therefore remains locked; the remaining problem is interaction consumption and calibration rather than another approximation of the current pair target.

The complementary expansion study reaches the same operational decision by a different route. Dense selected-only search gains 0.89 support-span points on held-out test at 12.65% extra source tokens, but admits 95.3% distractors and lowers 1.7B answer F1 when materialized independently. Gold-support expansion also lowers F1. The 512-token factorial shows that retrieval-linked attention repairs 84.1% of the expansion loss, but its interaction interval includes zero. At 1,024 tokens, pair and boundary attention recover 37.1% and 41.8% of the packed F1 deficit with positive paired intervals, yet neither reaches packed F1 and official-score recovery remains weak. Qwen scale points and Llama-8B do not share a consistent effect. Paper 3.3 therefore establishes auditable sparse interaction and expansion candidates, not a deployable cross-document controller. The SDK default remains no expansion, oracle access remains experiment-only, and learned or recursive policies remain locked.

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A SDK Qualification and Runtime Contract

Paper 3.3 reuses the canonical Paper 4.5/Paper 3.2 **ContextRecord** and storage metadata. Interaction is request-local and must not mutate persistent source records. The runtime sequence is

ContextRecord $\rightarrow$ **SelectionReceipt** $\rightarrow$ **InteractionPlan** $\rightarrow$ **RequestLocalComposition**.

An **InteractionPlan** records selected record pairs, token spans, layers/heads, budget, policy revision, and model/runtime compatibility.

Table 12: Cross-document expansion SDK qualification.

Criterion	Result	Gate
Pinned deterministic policy and immutable receipt	BGE revisions and cache digest recorded	Pass
No test-set tuning	Validation-selected policy replayed unchanged	Pass
At most 20% extra native K/V	12.65% source-token fraction on test	Pass
At least 50% packed-gap F1 recovery	No positive packed gap; expansion lowers 1.7B F1	Fail
No meaningful quality regression	Negative 1.7B mean; inconsistent ladder	Fail
Independent-seed stability	Bootstrap resampling only, not five model runs	Not met
Bounded production latency	370.3 ms preparation plus 249.5 ms exhaustive search	Not met
Two sizes and cross-family check	Qwen 1.7B/4B/8B and Llama 8B	Pass
Authorization, fallback, and audit receipt	Fail-closed runtime tests	Pass

The full expansion-policy grid, receipt fields, and timing decomposition are retained in the publication artifacts. They support reproducibility but are not part of the mechanism claim.

B Interventional Replay Procedure

For each frozen selection, the implementation performs the following steps:

1. Encode the concatenated records with ordinary full-causal attention, observing only causal cross-record probabilities into a compressed $[L, H, |E|]$ graph.
2. Encode the same tokens with every cross-record edge blocked. This is the base mask to which sparse plans add visibility.

3. For each document-pair or layer group g , replay the full graph except g , score the gold answer without decoding, and record $u_{\text{NLL}}(g)$ and $u_{\text{JS}}(g)$.
4. Order groups by one measured utility. Within a group, order physical edges by teacher attention, or multiply non-negative group NLL utility by edge attention for the combined proxy.
5. Replay nested prefixes at each physical-edge budget. Generate and teacher-force the answer from the same immutable selection and query.
6. Verify that the empty plan matches the blocked control and that the full plan matches the instrumented packed teacher exactly.

The implementation obtains only the prefix needed by the largest sparse budget through deterministic partition selection. It does not sort all tens of millions of edges for each target. This optimization changes oracle-analysis latency, not the selected set or the dense masked host execution used to test quality.

C Artifact Schema

Each run should persist:

- dataset/query IDs and seed;
- candidate and selection receipts;
- persistent record URIs/versions;
- document order;
- teacher interaction graph digest;
- oracle/learned interaction-plan digest;
- leave-one-group-out NLL and first-step JS utility;
- selected pairs/layers/heads/tokens;
- interaction budget;
- training checkpoint/policy revision;
- precision/runtime metadata;
- task/distribution/system metrics;
- paired bootstrap effects against the packed teacher;
- reranker execution device and frozen-split digest;
- git commit and hardware.

D Reproduction Map

The physical graph, plan, and localization implementation is in `src/prahf/sparse_crossdoc.py`. The host-preserving observer and sparse mask seam is in `src/prahf/rag_mlx_native.py`. The complete frozen retrieval and oracle sweep is in `experiments/paper3_3_sparse_crossdoc/run_oracle_sparsity.py`; split generation is in the adjacent `freeze_splits.py`. Cross-record search, authorization, logical-interval deduplication, accounting, and qualification gates are implemented in `src/prahf/crossdoc_expansion.py`. Retrieval-linked pair and boundary plans are in `src/prahf/sparse_crossdoc.py`. The evidence frontier and native answer replay are respectively `experiments/paper3_3_crossdoc_expansion/run_expansion_frontier.py` and `experiments/paper3_3_crossdoc_expansion/run_expansion_generation.py`. Held-out retrieval, paired generation, and cross-model figures are reduced by the adjacent `summarize_expansion.py`, `summarize_generation.py`, and `aggregate_generation_ladder.py`; the region-layer selection/consumption audit is reduced independently by `summarize_region_layer.py`. These tools check or propagate the frozen selection-cache digest. Exact commands and evidence labels appear in the experiment and paper README files.